2. Show that the firing-rate distribution that maximizes the entropy when the mean of the firing rate is held fixed is an exponential, and compute its entropy for a fixed resolution Δr. Assume that the firing rate can fall anywhere in the range from 0 to ∞. Use Lagrange multipliers (see the Mathematical Appendix) to constrain the integral of $p[r]$ to 1 and the integral of $p[r]r$ to the fixed average firing rate.

Condition: 1) $\int_r dr\, p(r) = 1$.

2) $\int_r dr\, r\, p(r) = \mu$.

$$L(p(r), \lambda_0, \lambda_1) = -\int_r dr\, p(r) \log p(r) - \lambda_0 \left(\int_r dr\, p(r) - 1 \right)$$

$$- \lambda_1 \left(\int_r dr\, r\, p(r) - \mu \right)$$

$f(x)\Big|_{x=a}^{x=b} = f(b) - f(a)$.

$$\frac{\partial L}{\partial p(r)} = -\log p(r) - 1 - \lambda_0 - r\lambda_1 = 0$$

$$\frac{\partial L}{\partial \lambda_0} = -\int_r dr\, p(r) + 1 = 0$$

$$\frac{\partial L}{\partial \lambda_1} = -\int_r dr\, r\, p(r) + \mu = 0.$$

Then $\log p(r) = -1 - \lambda_0 - r\lambda_1$.

$\therefore p(r) = \exp(1 - \lambda_0 - r\lambda_1)$.

Since $\int_0^\infty dr\, p(r) = 1$.

$$\int_0^\infty dr \exp(1 - \lambda_0 - r\lambda_1)$$

$$= \exp(1 - \lambda_0) \int_0^\infty dr \exp(-r\lambda_1) = 1.$$

$$\therefore \int_0^\infty dr \exp(-r\lambda_1) = \exp(+1 + \lambda_0).$$

$$-\frac{1}{\lambda_1} \exp(-r\lambda_1) \Big|_0^\infty = \frac{1}{\lambda_1}.$$

$$\therefore \frac{1}{\lambda_1} = \exp(+1 + \lambda_0).$$

$$\therefore \boxed{\lambda_1 = \exp(1 - \lambda_0)} \quad \lambda_1 = \exp(-1 - \lambda_0)$$

Also. $\int_0^\infty dr\, r \cdot p(r) = \mu$.

$$\int_0^\infty dr\, r \cdot \exp(-\lambda_0 - r\lambda_1) = \mu.$$

$$\exp(-\lambda_0) \int_0^\infty dr\, r \exp(-r\lambda_1) = \mu.$$

$$\lambda_1 \int_0^\infty dr\, r \exp(-r\lambda_1) = \mu.$$

$$\cancel{\lambda_1}\left[-\frac{1}{\cancel{\lambda_1}} \exp(-r\lambda_1)\left(r + \frac{1}{\lambda_1}\right)\right]_0^\infty = \mu.$$

$$\frac{1}{\lambda_1} = \mu. \quad \therefore \boxed{\lambda_1 = \frac{1}{\mu}}$$

$$\exp(-\lambda_0) = \frac{1}{\mu} \rightarrow -(-\lambda_0) = \ln\left(\frac{1}{\mu}\right)$$

$$\therefore \boxed{\lambda_0 = 1 - \ln\left(\frac{1}{\mu}\right)} \quad \boxed{\lambda_0 = -(-\ln\left(\frac{1}{\mu}\right)}$$

$$p(r) = \exp(-1 - \lambda_0 - r\lambda_1)$$

$$= \exp\left(\cancel{-1} \cancel{+1} + \ln\left(\frac{1}{\mu}\right) - r \cdot \frac{1}{\mu}\right),$$

$$= \frac{1}{\mu} \exp\left(-\frac{r}{\mu}\right) \quad (r \in [0, \infty)) \quad \blacksquare$$

3. Show that the distribution that maximizes the entropy when the mean and variance of the firing rate are held fixed is a Gaussian, and compute its entropy for a fixed resolution Δr. To simplify the mathematics, allow the firing rate to take any value between $-\infty$ and $+\infty$. Use Lagrange multipliers (see the Mathematical Appendix) to constrain the integral of $p[r]$ to 1, the integral of $p[r]r$ to the fixed average firing rate $\langle r \rangle$, and the integral of $p[r](r - \langle r \rangle)^2$ to the fixed variance.

Condition : 1) $\int_r dr\, p(r) = 1$.

2) $\int_r dr\, p(r)(r-\mu)^2 = \sigma^2$.

$$L(p(r), \lambda_0, \lambda_1) = -\int dr\, p(r) \log p(r) - \lambda_0 \int_r dr\, p(r) - 1)$$

$$- \lambda_1 \left(\int_r dr\, p(r)(r-\mu)^2 - \sigma^2 \right)$$

$$\frac{\partial L}{\partial p(r)} = -\log p(r) + 1 - \lambda_0 - \lambda_1 (x-\mu)^2$$

$$\therefore p(r) = \exp(-1 - \lambda_0 - \lambda_1 (x-\mu)^2).$$

Since, $\int_{-\infty}^{\infty} dr\, p(r) = 1$.

$$\int dr \exp\left(-1-\lambda_0-\lambda_1(r-\mu)^2\right) = 1.$$

$$\Leftrightarrow \exp(-1-\lambda_0)\int_{-\infty}^{\infty} dr \exp\left(-\lambda_1(r-\mu)^2\right) = 1$$

Let $r-\mu = z \rightarrow dr = dz$.

$$\exp(-1-\lambda_0)\int_{-\infty}^{\infty} dz \exp\left(-\lambda_1 z^2\right) = 1$$

$$\boxed{\exp(+1+\lambda_0) = \int_{-\infty}^{\infty} dz \exp\left(-\lambda_1 z^2\right) = \sqrt{\frac{\pi}{\lambda_1}}.}$$

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Since, $\int_r dr\, p(r)(r-\mu)^2 = \sigma^2$.

$$\Leftrightarrow \int_r dr \exp\left(-1-\lambda_0-\lambda_1(r-\mu)^2\right)\cdot(r-\mu)^2 = \sigma^2.$$

Let $r-\mu = z \rightarrow dr = dz$.

$$\exp(-1-\lambda_0)\int_{-\infty}^{\infty} dz \exp(-\lambda_1 z^2)\cdot z^2 = \sigma^2.$$

$$\sigma^2 \exp(+1+\lambda_0) = \int_{-\infty}^{\infty} dz\, z^2 \cdot \exp(-\lambda_1 z^2).$$

$$= \frac{1}{2}\sqrt{\frac{\pi}{\lambda_1^3}}$$

$$= \frac{1}{2}\frac{1}{\lambda_1}\sqrt{\frac{\pi}{\lambda_1}}$$

$$\therefore \boxed{\sigma^2 2\lambda_1 \exp(+1+\lambda_0) = \sqrt{\frac{\pi}{\lambda_1}}}$$

Since $\exp(-1+\lambda_0) = \sqrt{\frac{\pi}{\lambda_1}}$,

$$\sigma^2 2\lambda_1 \cancel{\exp(+1+\lambda_0)} = \cancel{\exp(+1+\lambda_0)},$$

$$\sigma^2 \cdot 2\lambda_1 = 1.$$

$$\therefore \boxed{\lambda_1 = \frac{1}{2\sigma^2}}$$

Also, $$e^{\lambda_0 + 1} = \sqrt{\frac{\pi}{\lambda_1}} = \sqrt{2\pi\sigma^2}.$$

$$\therefore \lambda_0 + 1 = \ln\sqrt{2\pi\sigma^2}.$$

$$\therefore \boxed{\lambda_0 = \ln\sqrt{2\pi\sigma^2} - 1.}$$

$$\therefore p(r) = \exp\left(-1 - \lambda_0 - \lambda_1 (r-\mu)^2\right)$$

$$= \exp\left(\cancel{-1} - \ln\sqrt{2\pi\sigma^2} \cancel{+1} - \frac{1}{2\sigma^2}(r-\mu)^2\right)$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(r-\mu)^2}{2\sigma^2}\right) \quad \blacksquare$$